CYBERSECURITY LAB 7

Configuring snort rules

1. Change of ipvar from any connection to our local network

```
# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overriden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 10.0.2.0/24
```

2. Setting rules for alerting on receiving any ICMP packet

```
*local.rules ×
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.


alert icmp any any -> any any (msg:"ICMP packet found";sid:1000001;rev:1;)
```

3. Output when pinged and alert received for receiving any ICMP packet.

```
cybersec-server@ubuntu:~$ sudo service snort restart
 * Starting Network Intrusion Detection System  snort                    [ OK ]
cybersec-server@ubuntu:~$
```

```
cybersec-attacker@ubuntu:~$ ping 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=64 time=0.319 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=64 time=0.330 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=64 time=0.289 ms
64 bytes from 10.0.2.6: icmp_seq=4 ttl=64 time=0.344 ms
64 bytes from 10.0.2.6: icmp_seq=5 ttl=64 time=0.324 ms
64 bytes from 10.0.2.6: icmp_seq=6 ttl=64 time=0.377 ms
64 bytes from 10.0.2.6: icmp_seq=7 ttl=64 time=0.366 ms
64 bytes from 10.0.2.6: icmp_seq=8 ttl=64 time=0.384 ms
64 bytes from 10.0.2.6: icmp_seq=9 ttl=64 time=0.371 ms
64 bytes from 10.0.2.6: icmp_seq=10 ttl=64 time=0.346 ms
64 bytes from 10.0.2.6: icmp_seq=11 ttl=64 time=0.305 ms
64 bytes from 10.0.2.6: icmp_seq=12 ttl=64 time=0.303 ms
64 bytes from 10.0.2.6: icmp_seq=13 ttl=64 time=0.301 ms
```

```
[**] [1:1000001:1] ICMP packet found [**]
[Priority: 0]
10/04-22:03:02.008761 10.0.2.7 -> 10.0.2.6
ICMP TTL:64 TOS:0x0 ID:3896 IpLen:20 DgmLen:84 DF
Type:8  Code:0  ID:3368   Seq:27  ECHO

[**] [1:1000001:1] ICMP packet found [**]
[Priority: 0]
10/04-22:03:02.008777 10.0.2.6 -> 10.0.2.7
ICMP TTL:64 TOS:0x0 ID:62839 IpLen:20 DgmLen:84
Type:0  Code:0  ID:3368  Seq:27  ECHO REPLY

cybersec-server@ubuntu:~$
```

4. Configuring snort on IDS and displaying alerts on console when pinged.

```
cybersec-server@ubuntu:~$ sudo snort -A console -q -c /etc/snort/snort.conf -i e
th0
10/04-22:05:36.033378  [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc
activity] [Priority: 3] {ICMP} 10.0.2.7 -> 10.0.2.6
10/04-22:05:36.033378  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {
ICMP} 10.0.2.7 -> 10.0.2.6
10/04-22:05:36.033378  [**] [1:384:5] ICMP PING [**] [Classification: Misc activ
ity] [Priority: 3] {ICMP} 10.0.2.7 -> 10.0.2.6
10/04-22:05:36.033396  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {
ICMP} 10.0.2.6 -> 10.0.2.7
10/04-22:05:36.033396  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc
 activity] [Priority: 3] {ICMP} 10.0.2.6 -> 10.0.2.7
10/04-22:05:37.033678  [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc
activity] [Priority: 3] {ICMP} 10.0.2.7 -> 10.0.2.6
10/04-22:05:37.033678  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {
ICMP} 10.0.2.7 -> 10.0.2.6
```

```
10/04-22:06:48.268995  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICM
P} 10.0.2.7 -> 10.0.2.6
10/04-22:06:48.269012  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {ICMP} 10.0.2.6 -> 10.0.
2.7
10/04-22:06:48.269012  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3
] {ICMP} 10.0.2.6 -> 10.0.2.7
10/04-22:06:49.269257  [**] [1:366:7] ICMP PING *NIX [**] [Classification: Misc activity] [Priority: 3]
 {ICMP} 10.0.2.7 -> 10.0.2.6
10/04-22:06:49.269257  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {ICMP} 10.0.2.7 -> 10.0.
2.6
10/04-22:06:49.269257  [**] [1:384:5] ICMP PING [**] [Classification: Misc activity] [Priority: 3] {ICM
P} 10.0.2.7 -> 10.0.2.6
10/04-22:06:49.269274  [**] [1:1000001:1] ICMP packet found [**] [Priority: 0] {ICMP} 10.0.2.6 -> 10.0.
2.7
10/04-22:06:49.269274  [**] [1:408:5] ICMP Echo Reply [**] [Classification: Misc activity] [Priority: 3
] {ICMP} 10.0.2.6 -> 10.0.2.7
```

5. Configuring specific rule to generate alert for web services

*local.rules ×

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.


alert icmp any any -> any any (msg:"ICMP packet found";sid:1000001;rev:1;)


alert tcp any any -> any any (msg:"web access request";sid:1000002;rev:1;)
```

```
[**] [1:1000002:1] web access request [**]
[Priority: 0]
10/04-22:21:29.218474 10.0.2.7:34356 -> 10.0.2.6:80
TCP TTL:64 TOS:0x0 ID:54931 IpLen:20 DgmLen:60 DF
******S* Seq: 0x11615A1C  Ack: 0x0  Win: 0x7210  TcpLen: 40
TCP Options (5) => MSS: 1460 SackOK TS: 494735 0 NOP WS: 7
```

6. Configuring snort rules for telnet connection

local.rules

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ---------------
# LOCAL RULES
# ---------------
# This file intentionally does not come with signatures.  Put your local
# additions here.


alert icmp any any -> any any (msg:"ICMP packet found";sid:1000001;rev:1;)


alert tcp any any -> any any (msg:"web access request";sid:1000002;rev:1;)

alert tcp any any -> any any (msg:"new telnet connection";sid:1000003;rev:1;)
```

```
[**] [1:1000003:1] new telnet connection [**]
[Priority: 0]
10/04-22:39:44.717836 10.0.2.7:45084 -> 10.0.2.6:23
TCP TTL:64 TOS:0x10 ID:36229 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x9FAF0DAC  Ack: 0x4304A711  Win: 0xED  TcpLen: 32
TCP Options (3) => NOP NOP TS: 768610 794567

[**] [1:1000003:1] new telnet connection [**]
[Priority: 0]
10/04-22:39:44.845208 10.0.2.7:45084 -> 10.0.2.6:23
TCP TTL:64 TOS:0x10 ID:36230 IpLen:20 DgmLen:52 DF
***A**** Seq: 0x9FAF0DAC  Ack: 0x4304A749  Win: 0xED  TcpLen: 32
TCP Options (3) => NOP NOP TS: 768642 794598

cybersec-server@ubuntu:~$
```